

HEPATIC ENCEPHALOPATHY



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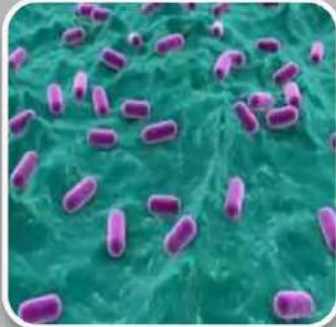
HEPATIC ENCEPHALOPATHY

Hepatic encephalopathy is one of the deadly complication of liver diseases, occurs due to profound liver failure and from accumulation of ammonia and other toxic metabolites in blood.

Hepatic coma is advanced stage of hepatic encephalopathy.



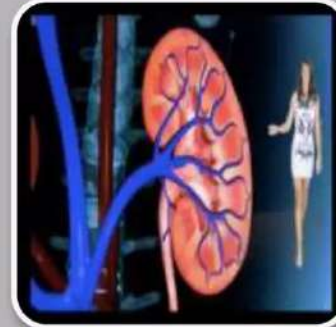
ETIOLOGY OF HEPATIC ENCEPHALOPATHY



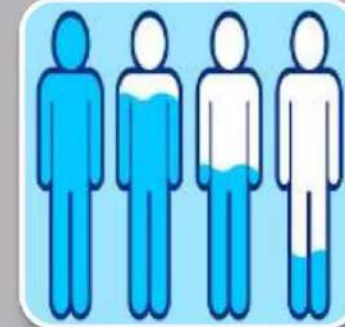
triggered
by a
build-up
of toxins
in the
blood
stream.



Infections
such as
Pneumonia



Kidney
problem.



Dehydration

CONT.



Constipation



Diuretics.



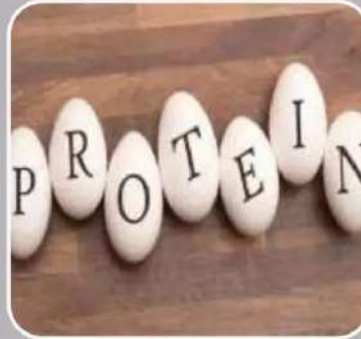
**Low
oxygen
levels
(hypoxia)**



**Recent
surgery
or
trauma.**



CONT.



Use of medications to suppress the immune system AND (such as barbiturates or benzodiazepine tranquilizers) that suppress the central nervous system.

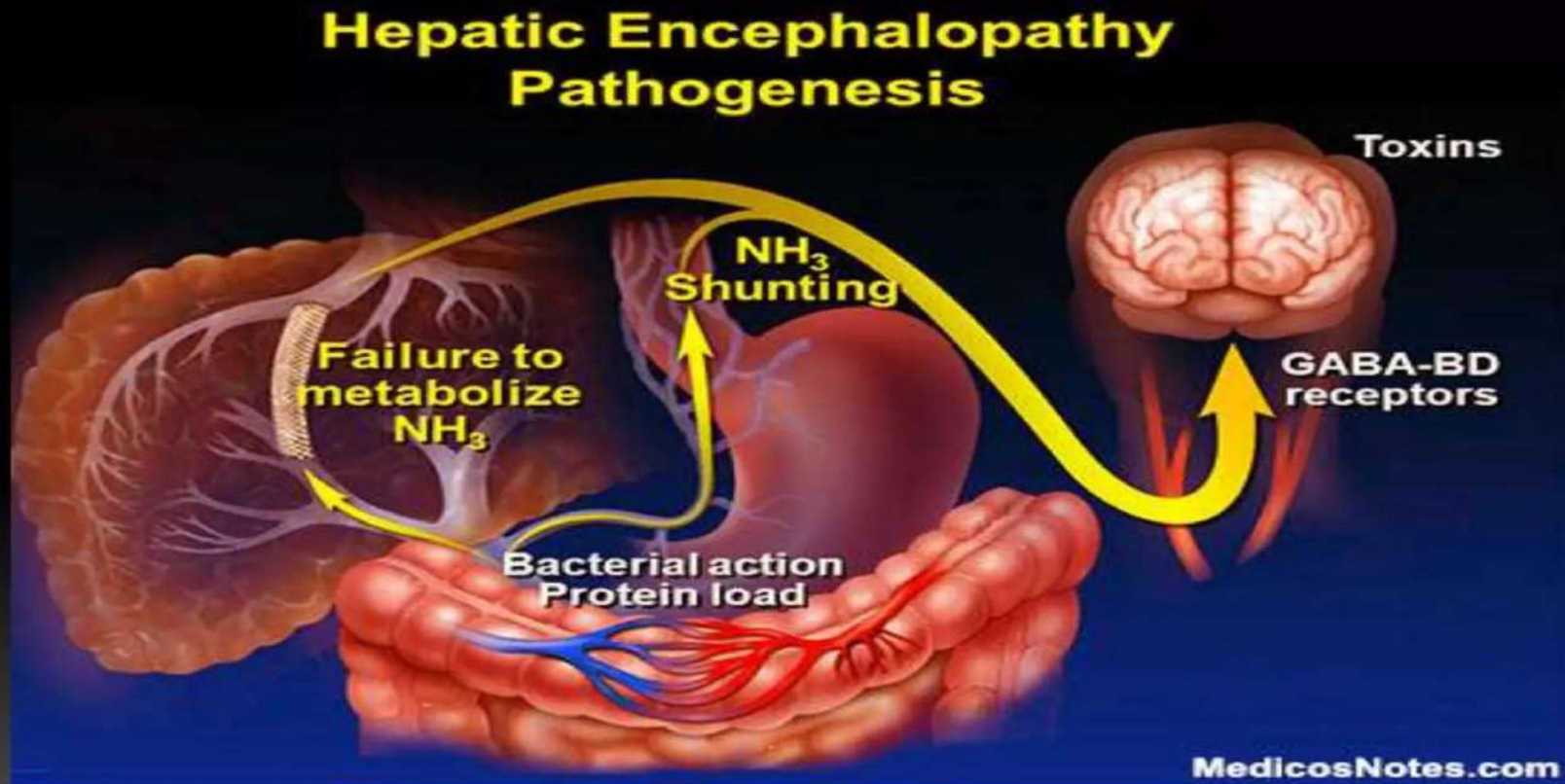
Increased protein intake.

Para-centesis

Electrolyte imbalance, especially a decrease in potassium.

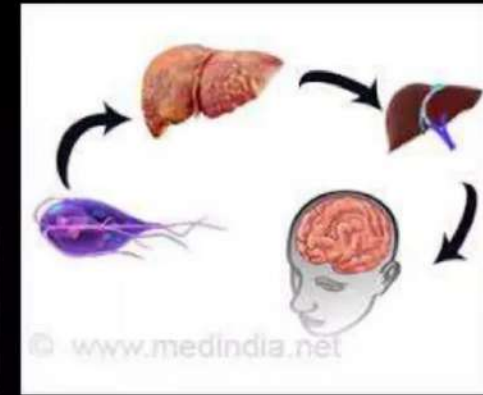


PATHOPHYSIOLOGY OF HEPATIC ENCEPHALOPATHY



PATHOPHYSIOLOGY

Damaged liver cells failed to detoxify and convert urea.

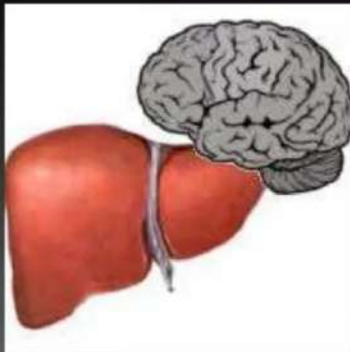


Ammonia constantly enter into blood stream from GI absorption, kidney and muscle cells.

Increased ammonia concentration in blood.

Brain dysfunction and damage.

Hepatic encephalopathy.



CLINICAL MANIFESTATION OF HEPATIC ENCEPHALOPATHY



CLINICAL MANIFESTATIONS



Minor mental changes.



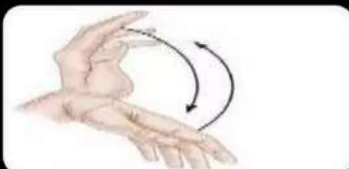
Motor disturbances.



Confused, experiences alteration in mood.



Altered sleep pattern. Insomnia.



Asterixis (flapping tremors of the hands).



CONT.



Changes in hand writing.



Fetor hepaticus (characteristic breath odour as a freshly mowed grass, acetone, old wine).



Disoriented patient with respect to time and place.



Coma. Seizures. Severe personality changes.

DIAGNOSTIC TESTS FOR HEPATIC ENCEPHALOPATHY



DIAGNOSTIC TESTS

Blood Tests: -A CBC may be used to check your RBC, WBC, and platelets. A low RBC count indicates blood loss and a lack of oxygen.



Blood tests may also be used to check the sodium, potassium, and ammonia levels in your blood. Having too much of these substances in the blood is a sign of impaired liver function.



CONT.



Imaging Tests: -An imaging test, such as a head CT scan or MRI, can check for bleeding in your head or abnormalities in your brain.

Liver Function Tests: -Liver function tests may be given to check for raised enzyme levels. An increase in enzymes indicates stress on the liver or liver damage.



MEDICAL MANAGEMENT OF HEPATIC ENCEPHALOPATHY



MEDICAL MANAGEMENT



Identify and treat precipitating causes:-

GI bleeding, increased dietary protein, constipation, infection, CNS depressant drugs and dehydration.

GI bleeding and increased protein intake may lead to increased bacterial formation of nitrogenous compounds that induce encephalopathy.





Reduce nitrogenous waste (Ammonia) in blood and bacteria in colon:- Neomycin and lactulose are given to reduce bacteria in the intestinal tract. Because it is not absorbed into the circulation, neomycin exerts a powerful effect on the intestinal bacteria that are responsible for ammonia production.

Maintain fluid volume balance: -with the accumulation of fluid in the abdominal area (ascites), bleeding, and decreased fluid intake, the client may experience a fluid volume deficit. This deficit, along with electrolyte imbalances that may occur, should be corrected. IV fluids are administered, carefully monitoring the quantity and rate of administration.

NURSING DIAGNOSIS OF HEPATIC ENCEPHALOPATHY



NURSING DIAGNOSIS

Ineffective therapeutic regimen management and ineffective family therapeutic regimen management related to reduction in protein in the diet and long term pharmacologic intervention with neomycin.

Deficit fluid volume related to bleeding, decreased intake, and ascites.

Risk for injury related to loss of protective mechanisms secondary to hepatic coma.

NURSING INTERVENTIONS



Promote low protein diet.



Monitor for GI haemorrhage.

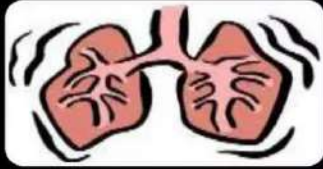


Encourage bowel cleansing.



Maintain intake output balance.

CONT.



Maintain patent airway.



Prevent infection.



Prevent ammonia toxicity and hypokalemia.



Avoid sedation.



Prevent complications of immobility.



**Thank
You!!!**